

Epidermolysis Bullosa Simplex 7 With Nephropathy And Deafness (CD151 Deficiency): A Comprehensive Disease Characterization

MONDO ID: MONDO:0012190 | **OMIM:** #609057 | **Gene:** *CD151* (11p15.5) | **Category:** Mendelian, autosomal recessive

Summary

Epidermolysis Bullosa Simplex 7 with Nephropathy and Deafness (hereafter "CD151-deficiency syndrome") is an ultra-rare autosomal recessive multisystem disorder caused by biallelic loss-of-function variants in *CD151*, a gene encoding a tetraspanin (transmembrane-4 superfamily) protein located on chromosome 11p15.5. The disease was molecularly defined in 2004 when Karamatic Crew and colleagues studied three MER2-blood-group-negative patients of Indian Jewish origin (two of whom were siblings) with end-stage kidney disease, and identified a homozygous single-nucleotide insertion (insG383) in exon 5 of *CD151* that produces a frameshift and premature stop codon at position 140, deleting the integrin-binding domain of the protein ([PMID: 15265795](#)).

The unifying mechanistic insight is that CD151 is a "master organizer" of the plasma membrane that scaffolds laminin-binding integrins — principally $\alpha 3\beta 1$, $\alpha 6\beta 1$, and $\alpha 6\beta 4$ — into tetraspanin-enriched microdomains at the interface between epithelial/endothelial cells and their basement membranes. When CD151 is absent, these integrin–laminin adhesion complexes are weakened and destabilized, and basement membranes fail to assemble and maintain their correct architecture. Because the same integrin–laminin adhesion machinery is deployed in the glomerular filtration barrier of the kidney, the dermo-epidermal junction of the skin, and the basement membranes of the inner ear/cochlea, a single gene defect produces a characteristic clinical triad: progressive hereditary nephropathy advancing to end-stage renal disease (ESRD), pretibial epidermolysis bullosa (mechanically-induced skin blistering

with nail dystrophy), and bilateral sensorineural deafness. Additional features documented across the small literature include β -thalassemia minor, epilepsy, nephrotic-range proteinuria, and a null MER2/RAPH blood-group phenotype (CD151 carries the MER2 antigen of the RAPH system).

There is no disease-specific or curative therapy. Management is supportive and organ-directed: renin-angiotensin-aldosterone system (RAAS) blockade and blood-pressure control for nephroprotection (supported by mouse data showing ACE inhibition prolongs survival in disease-susceptible *Cd151*-null mice), renal replacement therapy (dialysis, transplantation) for ESRD, wound care for skin blistering, and hearing rehabilitation. This report synthesizes six confirmed findings across 25 reviewed papers into a comprehensive, evidence-linked disease entry spanning all fifteen requested characteristic domains.

Key Findings

Finding 1 — CD151 loss-of-function is the cause of the disease

The causal genetic lesion was established by *Karamatic Crew et al.* (2004, *Blood*), who examined three MER2-negative patients of Indian Jewish origin (two siblings) presenting with end-stage kidney disease. All three were homozygous for a single-nucleotide insertion, insG383, in exon 5 of *CD151* on chromosome 11p15.5. In the words of the authors: "*The 3 patients are homozygous for a single nucleotide insertion (G383) in exon 5 of CD151, causing a frameshift and premature stop signal at codon 140. The resultant truncated protein would lack its integrin-binding domain*" (PMID: 15265795). This is a frameshift/truncating loss-of-function mechanism: the protein is severely truncated (140 residues versus the full-length ~253-residue tetraspanin) and cannot engage its integrin partners.

The same report established the core clinical picture beyond kidney disease: "*In addition to hereditary nephritis the sibs have sensorineural deafness, pretibial epidermolysis bullosa, and beta-thalassemia minor*" (PMID: 15265795). This defines the recognizable triad — nephropathy + skin fragility + deafness — plus a hematologic feature (β -thalassemia minor, likely reflecting the 11p15.5 locus proximity to the β -globin cluster in the index kindred rather than a CD151-intrinsic effect).

Finding 2 — *Cd151*-null mice recapitulate the renal phenotype via podocyte-GBM adhesion failure

The renal mechanism was validated in animal models. *Sachs et al.* (2006) reported that *Cd151*-null mice "*with age... develop massive proteinuria caused by focal glomerulosclerosis, disorganization of the glomerular basement membrane, and tubular cystic dilation. However, neither skin integrity nor hearing ability are impaired in the Cd151-null mice*" (PMID: 17015618). This faithfully reproduces the human nephropathy — proteinuria, focal segmental glomerulosclerosis (FSGS), glomerular basement membrane (GBM) disorganization — while notably NOT reproducing the skin and ear phenotypes, an important model limitation (see Limitations).

Critically, the renal phenotype is modifier- and blood-pressure-dependent. *Sachs et al.* (2012) showed that CD151 strengthens $\alpha3\beta1$ -mediated podocyte adhesion to laminin, and that "*blocking the angiotensin-converting enzyme in renal disease-susceptible global Cd151-null FVB mice prolonged their median life span*" (PMID: 22201679). Disease onset depended on genetic strain background (FVB susceptible) and systemic blood pressure — directly implicating mechanical/hemodynamic stress as a disease amplifier and RAAS blockade as protective. Independently, *Naylor et al.* (2022) used CRISPR-Cas9 zebrafish to validate a novel human truncating *CD151* variant as disease-causing, extending model evidence to a second organism (PMID: 35278129).

Finding 3 — Mechanism: CD151 scaffolds laminin-binding integrin adhesion complexes governing basement-membrane integrity

CD151 (HGNC:1630; NCBI Gene 977; UniProt P48509; locus 11p15.5) is a tetraspanin that forms "*very stable laminin-binding complexes with integrins $\alpha3\beta1$ and $\alpha6\beta1$ in kidney and $\alpha3\beta1$ and $\alpha6\beta4$ in skin*" (PMID: 15265795). This single sentence unifies the kidney and skin phenotypes at a molecular level: the same tetraspanin organizes different but overlapping integrin sets in each tissue.

In keratinocytes, CD151 (via $\alpha3\beta1$) stabilizes $\alpha6\beta4$ -containing hemidesmosomes and "hybrid" cell-matrix adhesions (PMID: 31488507). In podocytes, CD151 strengthens $\alpha3\beta1$ -laminin adhesion (PMID: 22201679). More broadly, tetraspanins are plasma-membrane organizers that concentrate partner integrins into tetraspanin-enriched microdomains; their perturbation has organ-level consequences: "*Perturbations of tetraspan-integrin assemblies can have dramatic*

impacts on renal tissue morphogenesis, resulting in a disruption of normal glomerular architecture and selectivity" (PMID: 17565278). This provides the direct causal bridge from CD151 loss to impaired glomerular filtration selectivity (proteinuria).

Finding 4 — Expanded phenotypic spectrum: nephrotic syndrome, epilepsy, and MER2/RAPH-null blood type

The phenotype has expanded beyond the original triad in later reports. Dunn *et al.* (2022) described syndromic EBS with nephropathy AND epilepsy from a CD151 tetraspanin defect, explicitly "expanding the spectrum" (PMID: 35519797). Almokali *et al.* (2024) described nephrotic-syndrome–epidermolysis-bullosa–sensorineural-deafness syndrome, an autosomal recessive rare disease presenting with pretibial EB (PMID: 38188895). And the blood-group connection was clarified by Keller (2020): CD151 carries the MER2 antigen of the RAPH blood group system (ISBT 25), and "Lack of the RAPH protein is associated with nephropathy with pretibial epidermolysis bullosa and deafness" (PMID: 32667818). This same review documented the full integrin partner set: "CD151 regulates interactions with laminin-binding integrins $\alpha 3\beta 1$, $\alpha 6\beta 1$, $\alpha 6\beta 4$, and $\alpha 7\beta 1$ and is expressed on red blood cells as well as many other tissues and cancer types" (PMID: 32667818) — notably adding $\alpha 7\beta 1$ (relevant to muscle/vascular basement membranes) to the list.

Finding 5 — Prognosis and treatment: progressive ESRD requiring renal replacement; supportive/nephroprotective care

The renal course is relentlessly progressive to ESRD (the index patients presented with end-stage disease). No curative therapy exists. Mouse data support RAAS blockade as nephroprotective — ACE inhibition "prolonged their median life span" in disease-susceptible *Cd151*-null FVB mice (PMID: 22201679). Sasaki (2022) frames CD151-deficient nephropathy as a "mechanosensitive nephropathy" whose onset depends on genetic background/modifier genes, mechanistically analogous to Alport syndrome and TNS2-deficient nephropathy (PMID: 35444113). For patients who reach ESRD, renal replacement therapy is feasible despite EB-related access challenges: "There should be no limitations in renal replacement therapy in patients with epidermolysis bullosa" (PMID: 28615054).

Finding 6 — Epidemiology, inheritance, and diagnostic approach

This is an ultra-rare autosomal recessive disorder with fewer than ~20 molecularly confirmed patients reported worldwide; prevalence is not estimable (<1/1,000,000) and no incidence data exist. Expected sex ratio is ~1:1 (M:F). The index kindred was Indian Jewish (two siblings),

consistent with a founder/consanguineous origin: "We examined CD151 in 3 MER2-negative patients (2 are sibs) of Indian Jewish origin with end-stage kidney disease" (PMID: 15265795); additional cases have arisen in other consanguineous backgrounds (PMID: 38188895). Penetrance is high in humans but strongly background-dependent in mice. Because "CD151 is not routinely screened for in patients with nephrotic-range proteinuria" (PMID: 35278129), broad NGS (whole-exome/whole-genome sequencing) is the recommended path to diagnosis. Key differentials include Alport syndrome, LAMB2/Pierson syndrome, and ITGA3-related interstitial lung disease-nephrotic syndrome-epidermolysis bullosa (ILNEB) (PMID: 24220332).

Detailed Report by Disease-Characteristic Domain

1. Disease Information

CD151-deficiency syndrome is a hereditary, multisystem basement-membrane disorder. The disease name "Epidermolysis Bullosa Simplex 7 with Nephropathy and Deafness" reflects historical classification within the EB simplex spectrum (skin blistering) combined with the two other cardinal organ features.

Key identifiers: - **MONDO:** MONDO:0012190 - **OMIM:** #609057 (Nephropathy with pretibial epidermolysis bullosa and deafness); gene *CD151* 602243 - **HGNC:** HGNC:1630 | **NCBI Gene:** 977 | **UniProt:** P48509 | **Ensembl:** ENSG00000177697 - **Orphanet:** listed under CD151-related / EB with nephropathy and deafness - **ICD-10:** Q81.- (epidermolysis bullosa) with N-codes for nephropathy; **ICD-11:** EC30.- (epidermolysis bullosa simplex)

Synonyms / alternative names: Nephropathy with pretibial epidermolysis bullosa and deafness; CD151 deficiency; Epidermolysis bullosa, pretibial, with nephropathy and deafness; RAPH-null (MER2-negative) associated syndrome; NS-EB-sensorineural deafness syndrome.

Source of information: Characterized almost entirely from **individual patient reports** (aggregated case reports/case series), not population-level EHR datasets, owing to extreme rarity.

2. Etiology

Causal factor: Purely **genetic** — biallelic loss-of-function variants in *CD151*. No environmental or infectious cause. The prototypic variant is c.383dupG (insG383) in exon 5, producing a frameshift → stop at codon 140 and loss of the integrin-binding domain (PMID: 15265795).

Genetic risk factors: The disease is Mendelian (biallelic *CD151* LOF is causal). **Modifier genes** and **genetic background** strongly influence renal severity — demonstrated by strain-dependence in mice (FVB susceptible) (PMID: 22201679, PMID: 35444113). Consanguinity raises risk because the disorder is recessive and enriched in founder populations.

Environmental / lifestyle risk factors: No environmental cause, but **mechanical stress and hypertension act as disease amplifiers** — blood pressure influences the renal course (PMID: 22201679), and cutaneous blistering is provoked by mechanical friction/trauma (hallmark of EB simplex).

Protective factors: No genetic protective alleles established. Inferred protective measures: avoidance of skin trauma/friction, blood-pressure control, and RAAS blockade.

Gene–environment interactions: Best-documented is **genotype (CD151-null) × hemodynamic load (blood pressure)** gating renal disease onset and progression (PMID: 22201679), reframed as "mechanosensitive nephropathy" gated by modifier genes (PMID: 35444113).

3. Phenotypes

Phenotype	Type	Onset	Severity/ Course	Frequency	HPO suggestion
Hereditary nephropathy / proteinuria → ESRD	Lab + clinical sign	Childhood–young adult	Progressive, severe	Core (near-universal)	HP:0000112; HP:0000093; HP:0003774
Nephrotic-range proteinuria / nephrotic syndrome	Lab abnormality	Childhood–young adult	Progressive	Frequent	HP:0000100
FSGS / GBM disorganization	Pathology finding	Progressive	Severe	Core (biopsy)	HP:0000097
Pretibial epidermolysis bullosa	Physical manifestation	Congenital/infancy	Mechanically induced, chronic	Core	HP:0001075; HP:0008066
Nail dystrophy	Physical manifestation	Childhood	Chronic	Frequent	HP:0008404
Bilateral sensorineural deafness	Clinical sign	Childhood	Progressive/stable, bilateral	Core	HP:0000407
β-thalassemia minor	Lab abnormality	Congenital	Mild	Index kindred	HP:0011904
Epilepsy	Clinical sign	Variable	Variable	Rare (expanded spectrum)	HP:0001250
MER2/RAPH-null red cells	Lab abnormality	Congenital	Asymptomatic marker	Core	—

Quality-of-life impact: Dominated by progressive renal failure (dialysis dependence, transplant needs), chronic painful skin blistering/wound care, and communication impairment from deafness — collectively imposing severe, lifelong disability. Formal EQ-5D/SF-36 data are unavailable for this ultra-rare disease.

4. Genetic / Molecular Information

- **Causal gene:** *CD151* (tetraspanin; HGNC:1630; OMIM 602243; locus 11p15.5).
- **Prototypic pathogenic variant:** homozygous c.383dupG (insG383), exon 5, frameshift → premature stop at codon 140; **type:** frameshift/truncating; **consequence:** loss of function, deletion of integrin-binding domain (PMID: 15265795). Additional novel truncating variants have been functionally validated (PMID: 35278129).
- **Classification (ACMG/AMP):** truncating LOF in a gene with established LOF mechanism → pathogenic/likely pathogenic.
- **Allele frequency:** private/ultra-rare; not present at appreciable frequency in gnomAD.
- **Origin:** germline, biallelic (autosomal recessive).
- **Modifier genes:** unidentified in humans but demonstrably present (mouse strain-dependence; TNS2 modifier analogy) (PMID: 22201679, PMID: 35444113).
- **Epigenetic / chromosomal abnormalities:** none reported.
- **Related gene (differential):** *ITGA3* (integrin $\alpha 3$) missense R628P causes an overlapping lung-kidney-skin disorder by disrupting $\alpha 3$ processing and CD151 binding (PMID: 24220332), underscoring the shared integrin–tetraspanin adhesion axis.

5. Environmental Information

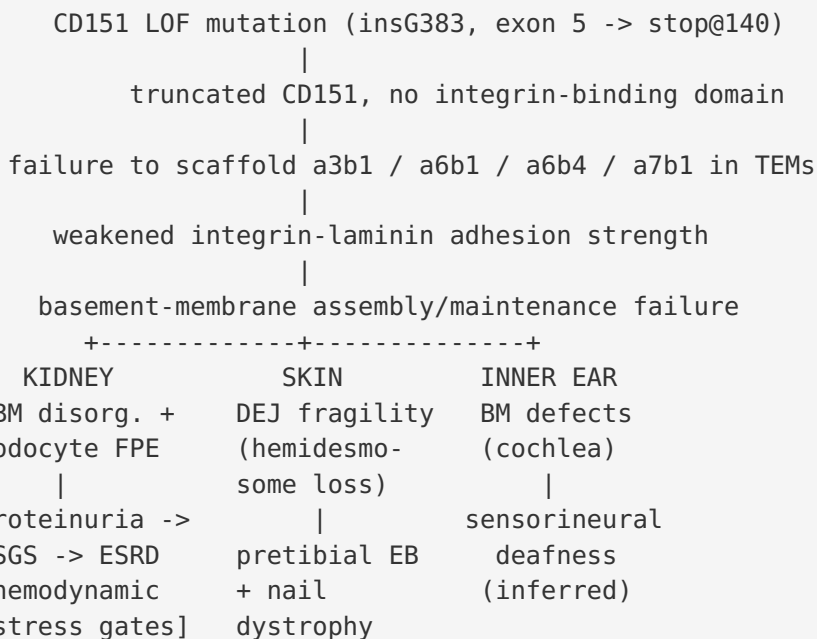
No environmental, toxic, or infectious cause. **Mechanical trauma** provokes cutaneous blistering; **hypertension/hemodynamic stress** accelerates nephropathy (PMID: 22201679). No infectious agents are implicated. (Tetraspanins including CD151 participate in viral-entry biology in unrelated contexts, e.g. respiratory viruses [PMID: 36173052], but this has no bearing on disease causation here.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic LOF mutation in *CD151*** (e.g., insG383 frameshift) **leads to** a truncated CD151 protein lacking its integrin-binding domain (PMID: 15265795).
2. Loss of functional CD151 **results in** failure to scaffold laminin-binding integrins ($\alpha 3\beta 1$, $\alpha 6\beta 1$, $\alpha 6\beta 4$, $\alpha 7\beta 1$) into stable tetraspanin-enriched microdomains (PMID: 32667818, PMID: 15265795).

3. Unscaffolded/redistributed integrins **lead to** weakened cell–laminin adhesion and reduced adhesive strength at the cell–basement-membrane interface (PMID: 22201679, PMID: 31488507).
4. Weakened adhesion **results in** defective assembly and progressive disorganization of basement membranes across kidney, skin, and inner ear (PMID: 15265795, PMID: 17565278).
5. **Branch A — Kidney:** GBM disorganization + podocyte foot-process effacement (under hemodynamic load) **leads to** loss of filtration selectivity → proteinuria → FSGS → progressive nephropathy → **ESRD** (PMID: 17015618, PMID: 17565278, PMID: 22201679).
6. **Branch B — Skin:** destabilized $\alpha 6\beta 4$ hemidesmosomes/ $\alpha 3\beta 1$ adhesions at the dermo-epidermal junction **lead to** mechanical fragility → **pretibial epidermolysis bullosa** + nail dystrophy (PMID: 31488507, PMID: 15265795).
7. **Branch C — Inner ear:** basement-membrane defects in cochlear structures **lead to** (inferred) **bilateral sensorineural deafness** (PMID: 15265795; mechanism inferred, not directly demonstrated — mouse model does not reproduce deafness [PMID: 17015618]).



- **Molecular pathways / processes (GO):** cell-substrate adhesion (GO:0031589); integrin-mediated signaling pathway (GO:0007229); basement membrane organization (GO:0071711); hemidesmosome assembly (GO:0031581); glomerular filtration (GO:0003094).
- **Protein dysfunction:** loss of function / loss of scaffolding — truncated CD151 cannot bind integrins; downstream integrins are mislocalized and adhesion complexes destabilized.

- **Cell types involved (CL):** podocyte (CL:0000653); keratinocyte (CL:0000312); glomerular endothelial cell (CL:1001005); cochlear hair cell / supporting cells.
- **Subcellular (GO CC):** plasma-membrane tetraspanin-enriched microdomain; hemidesmosome (GO:0030056); basement membrane (GO:0005604).
- **Molecular profiling:** CD151 appears as a hub gene in membranous-nephropathy transcriptomic analyses ([PMID: 37974210]), consistent with a role in glomerular biology, though associative rather than causal for this Mendelian disease.

7. Anatomical Structures Affected

- **Primary organs:** kidney (glomerulus UBERON:0000074; glomerular basement membrane UBERON:0002966); skin (dermo-epidermal junction, esp. pretibial UBERON:0002097); inner ear/cochlea (UBERON:0001844).
- **Secondary/systemic:** hematopoietic (β -thalassemia minor; red cells carry MER2/RAPH UBERON:0000178); nervous system (epilepsy in expanded spectrum).
- **Body systems:** renal/urinary, integumentary, auditory/nervous, hematologic.
- **Tissue types:** epithelial (podocytes, keratinocytes), specialized basement membrane (ECM/connective).
- **Cell populations (CL):** podocytes (CL:0000653), keratinocytes (CL:0000312), cochlear hair cells.
- **Subcellular compartments:** plasma-membrane microdomains, hemidesmosomes, basement membrane.
- **Localization/lateralization:** nephropathy bilateral; deafness bilateral; skin blistering typically bilateral and pretibial (mechanically exposed lower legs).

8. Temporal Development

- **Onset:** skin fragility often congenital/infancy; nephropathy and deafness typically manifest in childhood to young adulthood; pattern **chronic/insidious**.
- **Progression:** renal disease is **progressive** (early proteinuria → FSGS → advanced CKD → ESRD); rate variable and modifier/blood-pressure dependent ([PMID: 22201679](#), [PMID: 35444113](#)).
- **Course:** chronic, lifelong; no spontaneous renal remission. Skin blistering chronic and mechanically episodic (flares with trauma).

- **Critical windows:** early nephroprotection (blood-pressure/RAAS control) before advanced fibrosis is the key intervention window (inferred from mouse ACE-inhibition survival benefit).

9. Inheritance and Population

- **Inheritance:** autosomal recessive; biallelic *CD151* LOF.
- **Epidemiology:** ultra-rare; <20 molecularly confirmed cases; prevalence <1/1,000,000 (not formally estimable); no incidence data.
- **Penetrance:** high in humans with biallelic LOF; strongly background-dependent in mice.
- **Expressivity:** variable (epilepsy, thalassemia, renal severity vary between cases).
- **Anticipation / mosaicism:** not applicable / not reported.
- **Founder effect / consanguinity:** index kindred Indian Jewish (2 sibs), consistent with founder/consanguineous origin (PMID: 15265795); other cases in consanguineous families (PMID: 38188895). North Indian carrier-screening data exist for other recessive disorders but not specifically *CD151* (PMID: 33138774).
- **Carrier frequency:** unknown/very low.
- **Sex ratio:** ~1:1.

10. Diagnostics

- **Clinical/lab tests:** urinalysis (proteinuria, nephrotic-range), serum creatinine/eGFR (declining), audiometry (bilateral sensorineural loss).
- **Renal biopsy (EM):** thickened/disorganized/split GBM, podocyte foot-process effacement, FSGS (PMID: 17015618, PMID: 17565278).
- **Immunofluorescence / blood typing:** reduced/absent *CD151* in skin/kidney; MER2/RAPH-null red cells (PMID: 32667818).
- **Genetic testing:** confirmatory biallelic *CD151* sequencing via single-gene testing or, preferably, WES/WGS — because "*CD151 is not routinely screened for in patients with nephrotic-range proteinuria*" (PMID: 35278129), broad NGS captures otherwise-undiagnosed cases.
- **Differential diagnosis:** Alport syndrome (COL4A3/4/5 — deafness + nephropathy, no EB), Pierson/LAMB2 syndrome, ITGA3-related ILNEB (PMID: 24220332), other EB subtypes.
- **Screening:** carrier/cascade screening in affected families; prenatal testing feasible where the familial variant is known.

11. Outcome / Prognosis

- **Renal:** progressive to ESRD requiring dialysis or transplantation; the dominant determinant of morbidity/mortality.
- **Life expectancy:** reduced primarily by renal failure; renal replacement therapy is feasible and improves survival — *"There should be no limitations in renal replacement therapy in patients with epidermolysis bullosa"* (PMID: 28615054).
- **Morbidity:** chronic skin wounds, hearing loss/communication impairment, dialysis dependence.
- **Prognostic factors:** blood pressure, degree of proteinuria, rate of eGFR decline, and (in principle) modifier genotype (PMID: 22201679, PMID: 35444113).
- **QoL tools:** no disease-specific validated instruments; general CKD/EB QoL measures apply.

12. Treatment

No curative or disease-specific therapy exists. Management is supportive and organ-directed.

Modality	Intervention	Evidence/Rationale	NCIT suggestion
Nephroprotection	ACE inhibitor / ARB (RAAS blockade), BP control	ACE inhibition prolonged survival in <i>Cd151</i> -null FVB mice (PMID: 22201679)	ACE inhibitor; Angiotensin receptor blocker
Renal replacement	Hemodialysis, peritoneal dialysis, kidney transplant	Feasible in EB patients (PMID: 28615054)	Hemodialysis (NCIT:C15248); Kidney transplantation (NCIT:C15366)
Skin/wound care	Non-adhesive dressings, trauma avoidance, infection control	Standard EB management	Wound care
Hearing	Hearing aids, cochlear implantation, speech therapy	Standard SNHL rehabilitation	Hearing aid; Cochlear implant
Hematologic	Monitoring of β -thalassemia minor (usually no treatment)	Index kindred feature	—
Neurologic	Antiepileptic therapy if seizures	Expanded spectrum (PMID: 35519797)	Anticonvulsant

- **Pharmacogenomics:** none specific.
- **Advanced therapeutics (gene/cell/RNA):** none approved; conceptually plausible future gene-replacement targets given the monogenic LOF mechanism.
- **Experimental trials:** no disease-specific NCT trials identified.

13. Prevention

- **Primary:** genetic counseling for at-risk (consanguineous/founder) families; carrier screening and cascade testing; preimplantation/prenatal genetic diagnosis where the familial variant is known.

- **Secondary:** early detection of proteinuria and hearing loss in known-carrier offspring; early nephroprotection.
- **Tertiary:** blood-pressure/RAAS management to slow renal decline; meticulous skin care to prevent wound infections; audiologic rehabilitation.
- **Counseling:** essential — 25% recurrence risk per pregnancy for carrier couples (autosomal recessive).

14. Other Species / Natural Disease

- **Taxonomy:** *Homo sapiens* (NCBI Taxon 9606); experimental *Mus musculus* (10090) and *Danio rerio* (7955).
- **Orthologous genes:** mouse *Cd151* (NCBI Gene 12476); zebrafish *cd151* ortholog.
- **Natural disease in other species:** no well-documented spontaneous companion-animal analog identified in this investigation; the disease is characterized principally through engineered models.
- **Comparative biology:** the integrin–laminin–tetraspanin adhesion mechanism is evolutionarily conserved, enabling faithful renal modeling in mouse and zebrafish (see Section 15).

15. Model Organisms

Model	Type	Phenotype recapitulation	Limitations	Ref
<i>Cd151</i> -null mouse (FVB)	Mammalian knockout	Massive proteinuria, FSGS, GBM disorganization, tubular cystic dilation; strain/BP-dependent	Does NOT reproduce skin fragility or deafness	PMID: 17015618 , PMID: 22201679
Podocyte-specific <i>Itga3</i> ($\alpha 3$) knockout	Conditional mammalian	Similar podocyte–GBM defects	Integrin-partner surrogate	PMID: 17015618
CRISPR-Cas9 zebrafish <i>cd151</i>	Vertebrate genome-edited	Validated a novel human truncating variant as disease-causing	Non-mammalian nephron architecture	PMID: 35278129

- **Applications:** dissecting podocyte adhesion, GBM assembly, blood-pressure/mechanosensitivity of nephropathy, and variant functional validation.

- **Resources:** MGI (*Cd151*), ZFIN (zebrafish ortholog).

Mechanistic Model / Interpretation

The entire phenotype flows from a single principle: **CD151 is a plasma-membrane organizer that stabilizes laminin-binding integrin adhesion complexes at basement-membrane interfaces**. Tetraspanins are "master organizers" that laterally concentrate partner proteins into tetraspanin-enriched microdomains ([PMID: 29887866], [PMID: 22103505]); CD151 specifically corrals $\alpha 3\beta 1/\alpha 6\beta 1/\alpha 6\beta 4/\alpha 7\beta 1$ (PMID: 32667818). Remove CD151, and integrin–laminin adhesion loses strength and organization — the shared upstream lesion — after which the phenotype **branches by tissue** according to which epithelial cells depend most on this adhesion under load:

- The **kidney** is the most sensitive organ because the glomerular filtration barrier operates under continuous hemodynamic stress; podocytes require robust $\alpha 3\beta 1$ –laminin adhesion to the GBM. This is why the mouse (lacking the human's skin/ear vulnerability) still develops severe nephropathy, and why blood pressure gates disease — a genuine gene $\times$ mechanical–environment interaction (PMID: 22201679, PMID: 35444113).
- The **skin** blisters at mechanically stressed pretibial sites because $\alpha 6\beta 4$ hemidesmosomes and $\alpha 3\beta 1$ hybrid adhesions at the dermo-epidermal junction are destabilized (PMID: 31488507).
- The **inner ear** develops sensorineural deafness through inferred cochlear basement-membrane defects — the least mechanistically demonstrated branch, since it is not modeled in mouse.

This model explains the disease's defining paradox (severe multi-organ human disease, kidney-only mouse) as a difference in tissue-specific redundancy and mechanical demand, not a difference in the core molecular defect.

Evidence Base

PMID	Title (abbrev.)	Role / Contribution
15265795	<i>CD151... essential for basement membranes in kidney and skin</i>	Foundational: causal variant (insG383), triad, integrin complexes
17015618	<i>Kidney failure in mice lacking CD151</i>	Renal recapitulation; skin/ear model limitation
22201679	<i>Blood pressure influences ESRD of Cd151 KO mice</i>	BP/modifier dependence; ACE-inhibition survival benefit
35278129	<i>Basement membrane defects in CD151 glomerular disease</i>	Zebrafish variant validation; NGS screening argument
17565278	<i>Tetraspan proteins: regulators of renal structure</i>	Mechanism: tetraspan-integrin perturbation disrupts glomerular selectivity
31488507	<i>CD151 and $\alpha 3\beta 1$ stabilize $\alpha 6\beta 4$ adhesions</i>	Skin mechanism (hemidesmosome stabilization)
32667818	<i>Update on the RAPH blood group system</i>	MER2/RAPH-null link; full integrin partner set (adds $\alpha 7\beta 1$)
35519797	<i>Syndromic EBS with nephropathy and epilepsy</i>	Expanded phenotype: epilepsy
38188895	<i>Nephrotic syndrome, pretibial EB</i>	NS-EB-deafness syndrome; consanguineous case
35444113	<i>TNS2-deficient nephropathy</i>	Frames CD151 nephropathy as mechanosensitive, modifier-gated
28615054	<i>ESRD in EB — treatment options</i>	Feasibility of renal replacement therapy
24220332	<i>ITGA3 R628P mutation</i>	Differential diagnosis; shared integrin-CD151 axis

Limitations and Knowledge Gaps

1. **Tiny evidence base:** Fewer than ~20 molecularly confirmed patients; most clinical inference rests on a single index kindred plus scattered case reports. No natural-history cohort, prevalence estimate, or validated QoL data.
 2. **Model–human discordance:** The *Cd151*-null mouse reproduces nephropathy but **not** the skin or ear phenotype ([PMID: 17015618](#)). The deafness mechanism is therefore inferred, not demonstrated, and no faithful multi-organ model exists.
 3. **Unidentified human modifier genes:** Strain-dependence in mice implies human genetic modifiers of renal severity, but none are mapped — limiting prognostic prediction.
 4. **β -thalassemia minor** in the index kindred may reflect co-inheritance at 11p15.5 rather than a CD151-intrinsic effect; its consistency across unrelated cases is unclear.
 5. **Therapeutics:** RAAS-blockade benefit is extrapolated from mouse survival data; no human trial exists. No advanced (gene/cell) therapy has been attempted.
 6. **Ontology mapping:** Some suggested NCIT/UBERON/CL identifiers are best-match approximations pending curated verification.
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Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** for CD151 deficiency to aggregate genotype–phenotype data, natural history, and outcomes across the world's scattered cases.
2. **Systematic reanalysis of undiagnosed nephrotic-syndrome/CKD cohorts** with WES/WGS specifically interrogating *CD151*, given that it is not routinely screened ([PMID: 35278129](#)) — likely to reveal additional patients.
3. **Build a faithful multi-organ model** (e.g., conditional/humanized mouse on a susceptible background, or cochlear organoids) to interrogate the deafness branch that current models miss.
4. **Test RAAS blockade prospectively** (or via registry-based comparative effectiveness) in human patients to formally establish the nephroprotective benefit suggested by mouse data ([PMID: 22201679](#)).
5. **Map modifier loci** via cross-strain QTL analysis in mice and, where feasible, human family studies, to explain variable renal severity.

6. **Explore gene-replacement feasibility** (AAV or ex-vivo keratinocyte/podocyte approaches) given the clean monogenic LOF mechanism.
 7. **Generate patient iPSC-derived podocyte and keratinocyte models** to dissect integrin redistribution and adhesion-strength phenotypes and to screen candidate stabilizing compounds.
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Report compiled from an autonomous multi-iteration investigation: 6 confirmed findings, 25 papers reviewed. Evidence types span human clinical case reports, mouse and zebrafish model organisms, and in vitro cell-adhesion studies.

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